

Wireless Fundamentals (Part-1)

- Topics covered:
 - Radio Frequency (RF)
 - Wi-Fi Standards
 - Wireless LAN Fundamentals

Radio Frequency

Introduction to RF and Wi-Fi

RF (Radio Frequency) is a range of electromagnetic wave frequencies assigned for various purposes, including:

1. AM radio
2. FM radio
3. Microwaves
4. Radar
5. Wi-Fi

Wireless LANs use Wi-Fi, which is the most important RF application for this section.

Wireless LAN standards are defined in IEEE 802.11, just as Ethernet standards are defined in IEEE 802.3.

IEEE 802.11 and Wi-Fi Alliance

IEEE Standards:

1. Wired Ethernet LANs → **IEEE 802.3**
2. Wireless LANs → **IEEE 802.11**

Wi-Fi Alliance:

1. Wi-Fi is a trademark of the **Wi-Fi Alliance**.
2. The Wi-Fi Alliance is not directly connected to the IEEE.

It tests and certifies devices for:

- 802.11 standards compliance
- Interoperability with other devices

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Wi-Fi Certified Mark



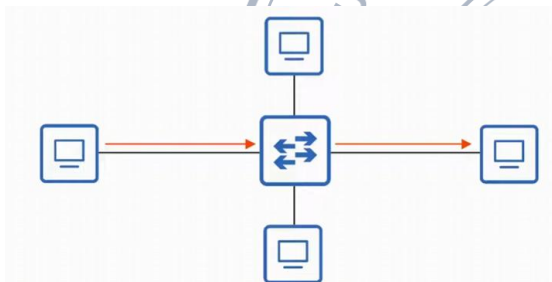
- Devices certified by the Wi-Fi Alliance can display the **Wi-Fi Certified** mark.
- Although Wi-Fi Alliance certifies devices for compliance with 802.11 standards, "Wi-Fi" is not technically the correct term for 802.11 WLANs.
- However, Wi-Fi has become the common term used for 802.11 wireless LANs.

Wireless LAN Fundamentals

Wireless LANs differ from wired LANs in several ways.

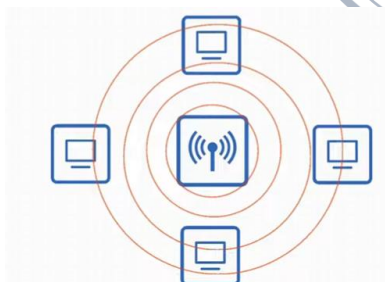
(All Devices Receive the Transmission)

Ethernet Switch



Other devices do not receive the frame.

Wireless LAN



1. When a wireless device transmits a frame, all wireless devices within range can receive it.
2. Unlike wired networks, the signal is not contained inside a cable.
3. The signal radiates outward as electromagnetic waves.

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Effects of Wireless Transmission

1. Data Privacy Concerns

Because all devices within range can receive transmissions:

- Data privacy becomes a greater concern.
- In wired LANs, data is usually not encrypted within the LAN.
- In wireless LANs, encryption is very important.
- Without encryption, any device within range can access transmitted data.

2. Collisions

Wireless devices using the same channel at the same time can cause collisions.

To reduce this issue, wireless LANs use:

CSMA/CA (Carrier Sense Multiple Access with Collision Avoidance)

Comparison:

Technology Purpose

CSMA/CD Detects and recovers from collisions in wired networks

CSMA/CA Avoids collisions before they occur in wireless networks

Basic Idea: A device waits until other devices stop transmitting before sending its own data.

CSMA/CA Flow



Process:

1. Assemble the frame.
2. Listen to determine whether the channel is free.
3. If the channel is busy: Wait for a random period and then Listen again.
4. If the channel is free: Transmit the frame.

Additional Information

An optional feature exists where:

1. Sender transmits an **RTS** (Request To Send).
2. Receiver responds with a **CTS** (Clear To Send).
3. Data is then transmitted.

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This is additional information and not required for CCNA.

Remember: Wireless LANs use CSMA/CA to avoid collisions.

Wireless Communication Regulations

Wireless communication is regulated by:

1. International bodies
2. National bodies

Important points:

- Devices cannot transmit on any channel they want.
- Allowed channels can differ between countries.
- IEEE 802.11 specifies frequencies that can be used for wireless LANs.
- Wireless devices are designed to operate on those frequencies.

Wireless Signal Coverage

Wireless networks require consideration of signal coverage.

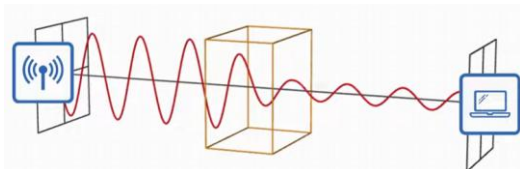
Factors affecting signal travel:

1. Absorption
2. Reflection
3. Refraction
4. Diffraction
5. Scattering

Absorption

Definition: Absorption occurs when a wireless signal passes through a material and part of the signal is converted into heat.

Result: Original signal becomes weaker.



Effect:

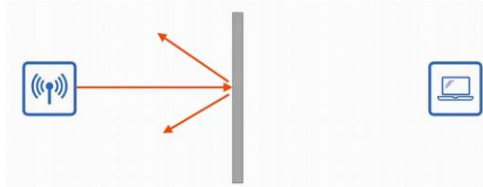
1. The wall absorbs part of the signal.
2. The laptop receives a weaker signal.

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Reflection

Definition: Reflection occurs when a signal bounces off a material.

Example material: Metal



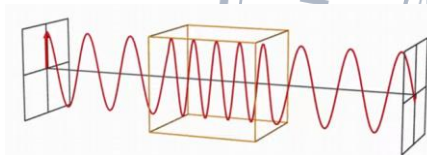
Effect:

1. Much of the signal bounces away.
2. Very little passes through the metal.
3. The receiving device gets a poor signal.

Refraction

Definition: Refraction occurs when a wave enters a medium where it travels at a different speed.

Examples: Glass, Water

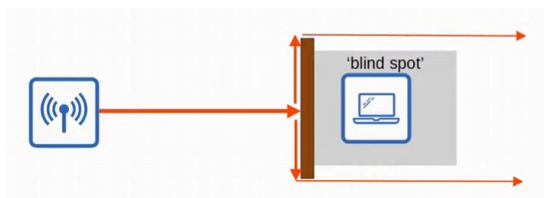


Effect

1. The wave bends when entering the new medium.
2. The change in speed causes the bending.

Diffraction

Definition: Diffraction occurs when a wave encounters an obstacle and travels around it.



Effect

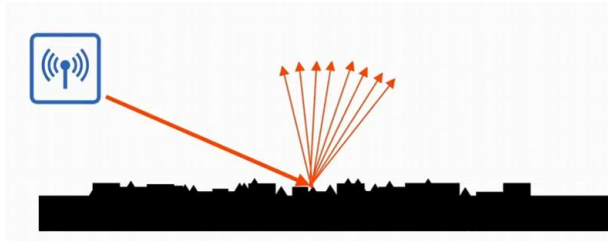
1. Signals can travel around obstacles to some degree.
2. Areas behind obstacles may become blind spots.
3. Devices in blind spots may not receive sufficient signal.

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Scattering

Definition: Scattering occurs when a material causes a signal to spread in many directions.

Causes include: Dust, Smog, Uneven surfaces



Effect:

1. Signal energy spreads in multiple directions.
2. Signal quality is reduced.

Impact on Wireless Network Design

The following factors affect wireless signal quality:

1. Absorption
2. Reflection
3. Refraction
4. Diffraction
5. Scattering

When positioning wireless access points, all of these factors must be considered.

Interference

Another issue in wireless networks is interference.

Cause: Other devices using the same channels.

Example: A neighboring wireless LAN.

Result: Wireless performance can be affected by transmissions from nearby devices.

Radio Frequency (RF) and Electromagnetic Waves

How Wireless Signals Are Created

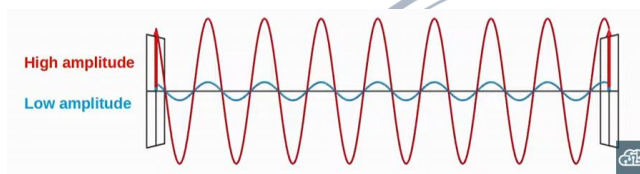
To send wireless signals:

1. The sender applies an **alternating current (AC)** to an antenna.
2. This creates electromagnetic fields.
3. These fields propagate outward as electromagnetic waves.

Electromagnetic waves can be measured in multiple ways, including: Amplitude, Frequency

Amplitude

Amplitude is the maximum strength of the electric and magnetic fields.



- The red wave has a higher amplitude.
- The blue wave has a lower amplitude.
- Both waves have the same frequency.

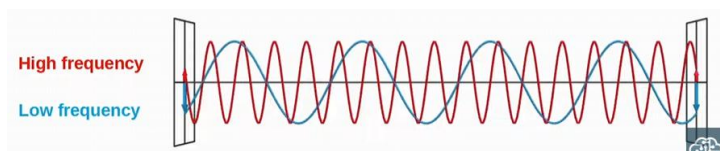
Frequency

Frequency measures the number of up/down cycles in a given unit of time.

Common Unit Hertz (Hz) = cycles per second

Other common units:

Unit	Meaning
Hertz (Hz)	Cycles per second
Kilohertz (kHz)	Thousands of cycles per second
Megahertz (MHz)	Millions of cycles per second
Gigahertz (GHz)	Billions of cycles per second
Terahertz (THz)	Trillions of cycles per second

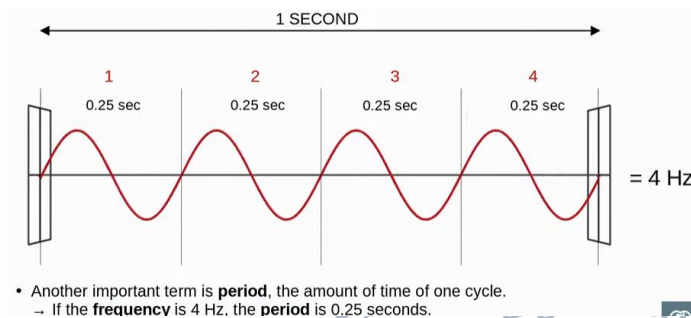


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- both waves have the same amplitude.
- The red wave has a higher frequency.
- The blue wave has a lower frequency.
- The red wave completes more cycles per second.

Frequency and Period

Consider the following wave representing one second.



Period: is the amount of time required for one cycle.

If: Frequency = 4 Hz

Then: Period = 0.25 seconds

Because: 4 cycles occur in 1 second

1 cycle = $\frac{1}{4}$ second = 0.25 seconds

Frequency Ranges

Visible Frequency Range: Approximately 400 THz to 790 THz

Radio Frequency (RF) Range: Approximately 30 Hz to 300 GHz

RF is used for many different purposes.

For wireless LANs, only small portions of this RF spectrum are used.

Wi-Fi Frequency Bands

IEEE 802.11 wireless LANs use portions of:

1. Ultra-High Frequency (UHF)
2. Super High Frequency (SHF)

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Main Wi-Fi Bands

2.4 GHz Band

Name: 2.4 GHz

Actual Range: 2.400 GHz – 2.4835 GHz

5 GHz Bands

Name: 5 GHz

Actual Range: 5.150 GHz – 5.825 GHz

This band is further divided into four smaller bands.

Remember only: 2.4 GHz band and 5 GHz band

Exact ranges are not necessary for the CCNA.

Divided into four smaller bands:

5.150 GHz to 5.250 GHz
5.250 GHz to 5.350 GHz
5.470 GHz to 5.725 GHz
5.725 GHz to 5.825 GHz

Characteristics of the 2.4 GHz Band

The 2.4 GHz band typically provides:

- Greater reach in open space
- Better penetration through obstacles such as walls

However:

- More devices use 2.4 GHz
- Interference is often greater than on 5 GHz

Modern devices commonly support both bands.

Wi-Fi 6 and 6 GHz

Wi-Fi 6 corresponds to: IEEE 802.11ax

Wi-Fi 6 expands the available spectrum to include: 6 GHz band

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Channels

Each frequency band is divided into multiple channels.

Devices are configured to transmit and receive on one or more channels.

A channel bonding feature can combine channels, but it is not required knowledge for the CCNA.

2.4 GHz Channels

The 2.4 GHz band contains multiple channels.

Each channel occupies: 22 MHz

The available channels vary by country.

Channel Table

Channel	F ₀ (MHz)	Frequency Range (MHz)	North America ^[B]	Japan ^[B]	Most of world ^{[B][9][10][11][12][13][14][15]}
1	2412	2401–2423	Yes	Yes	Yes
2	2417	2406–2428	Yes	Yes	Yes
3	2422	2411–2433	Yes	Yes	Yes
4	2427	2416–2438	Yes	Yes	Yes
5	2432	2421–2443	Yes	Yes	Yes
6	2437	2426–2448	Yes	Yes	Yes
7	2442	2431–2453	Yes	Yes	Yes
8	2447	2436–2458	Yes	Yes	Yes
9	2452	2441–2463	Yes	Yes	Yes
10	2457	2446–2468	Yes	Yes	Yes
11	2462	2451–2473	Yes	Yes	Yes
12	2467	2456–2478	No ^B except CAN	Yes	Yes
13	2472	2461–2483	No ^B	Yes	Yes
14	2484	2473–2495	No	11b only ^C	No

Channel 14 in Japan includes: 802.11b only

802.11b is an older and slower standard.

Channel Overlap

An important characteristic of 2.4 GHz channels is that they overlap.

Example: Channel 1: 2401 - 2423 MHz

This overlaps with: Channel 2, Channel 3, Channel 4, Channel 5

Because channels overlap, channel selection must be done carefully.

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Access Point Channel Planning

Single Access Point [AP]

1. Any channel can be used.
2. No neighbouring AP exists to cause interference.

Multiple Access Points [AP] [AP] [AP]

Adjacent APs should not use overlapping channels.

Reason:

- Reduces interference
- Improves performance
- Improves user experience

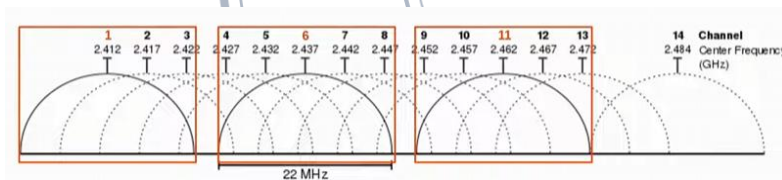
If overlapping channels are used:

- Performance decreases
- User experience worsens

Recommended Channels in 2.4 GHz

The recommended channels are: 1, 6, 11

Diagram Channel Grouping



These channels do not overlap.

Therefore:

1. AP 1 -> Channel 1
2. AP 2 -> Channel 6
3. AP 3 -> Channel 11

can operate without interfering with one another.

For the CCNA, remember: **1 / 6 / 11**

5 GHz Channels

The 5 GHz band consists of:

Non-overlapping channels

As a result:

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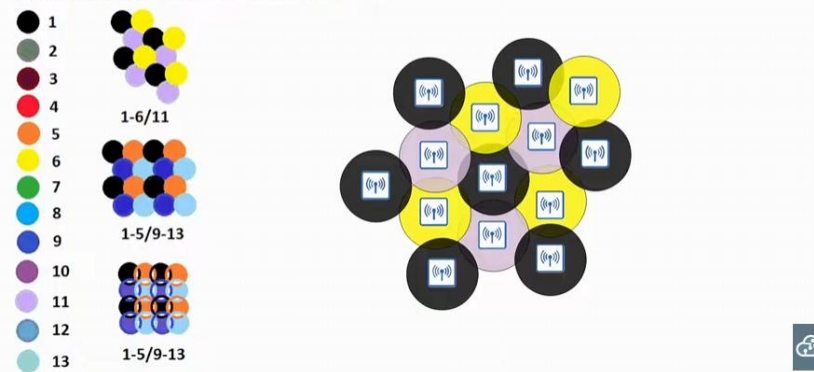
1. Avoiding interference is much easier.
2. Channel planning is simpler.

Honeycomb Access Point Design

Using channels 1, 6, and 11, APs can be arranged in a honeycomb pattern.

Conceptual Pattern:

Using channels 1, 6, and 11, you can place APs in a 'honeycomb' pattern to provide complete coverage of an area without interference between channels.



Key points:

- AP coverage areas overlap.
- Frequencies do not overlap.
- Complete area coverage is achieved.
- Interference is minimized.

This arrangement is useful when providing coverage across large areas.

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IEEE 802.11 Standards

Various Wi-Fi standards use different frequencies and provide different theoretical data rates.

Standard	Frequencies	Max Data Rate (theoretical)	Alternate Name
802.11	2.4 GHz	2 Mbps	
802.11b	2.4 GHz	11 Mbps	
802.11a	5 GHz	54 Mbps	
802.11g	2.4 GHz	54 Mbps	
802.11n	2.4 / 5 GHz	600 Mbps	'Wi-Fi 4'
802.11ac	5 GHz	6.93 Gbps	'Wi-Fi 5'
802.11ax	2.4 / 5 / 6 GHz	4*802.11ac	Wi-Fi 6'

Important Notes About Data Rates

The listed data rates are: Theoretical maximums

Actual performance is often lower for many reasons.

Wi-Fi Naming

The following alternate names are used:

Standard	Wi-Fi Name
802.11n	Wi-Fi 4
802.11ac	Wi-Fi 5
802.11ax	Wi-Fi 6

There are no official: Wi-Fi 1, Wi-Fi 2 and Wi-Fi 3

Device Support

802.11-enabled devices may support:

1. One standard
2. Several standards
3. All standards

Before purchasing a device, check which standards it supports.

Example An iPhone 10 support:

- 802.11a
- 802.11b
- 802.11g
- 802.11n
- 802.11ac

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802.11 Service Sets

802.11 defines different kinds of **service sets**, which are groups of wireless network devices.

There are three main types:

1. Independent Service Sets
2. Infrastructure Service Sets
3. Mesh Service Sets

All devices in a service set share the same **SSID (Service Set ID)**.

The SSID is a human-readable name used to identify the service set. It does not have to be unique, although it is generally best to configure unique SSIDs because users select wireless networks based on the SSID they see on their devices.

The SSID can be any human-readable name.

Example SSIDs Detected by a Device



Although these names are not especially human-friendly, the SSID can easily be changed to a simpler name.

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Independent Basic Service Set (IBSS)

An IBSS (Independent Basic Service Set) is a wireless network in which two or more wireless devices connect directly to each other without using an access point.

IBSS networks are also called **ad hoc networks**.

IBSS Structure



In an IBSS:

1. No AP is used.
2. Devices communicate directly.
3. It can be used for tasks such as file transfers.
4. It is not scalable beyond a few devices.
5. It is typically used only for limited purposes such as quick file transfers.

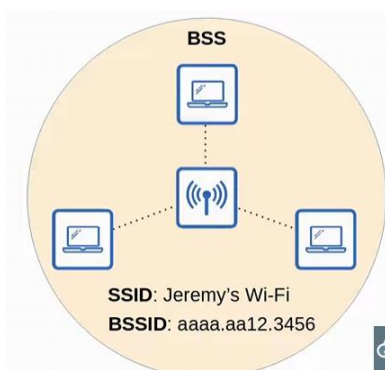
Basic Service Set (BSS)

A BSS (Basic Service Set) is an infrastructure service set.

In a BSS, clients connect through an AP rather than communicating directly with one another.

The AP provides the network infrastructure that allows wireless clients to communicate.

BSS Structure



A BSSID (Basic Service Set ID) uniquely identifies the AP.

Other APs may use the same SSID, but they cannot use the same BSSID.

The BSSID is the MAC address of the AP's radio, and MAC addresses are unique.

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Association

To become part of a BSS, wireless devices request association with the BSS.

After association, the devices are called: Clients and Stations

Basic Service Area (BSA)

Another important term is BSA (Basic Service Area).

The BSA is the physical area around an AP where its signal is usable.

Difference Between BSS and BSA

BSS: A group of devices connected through an AP.

BSA: The physical coverage area around the AP.

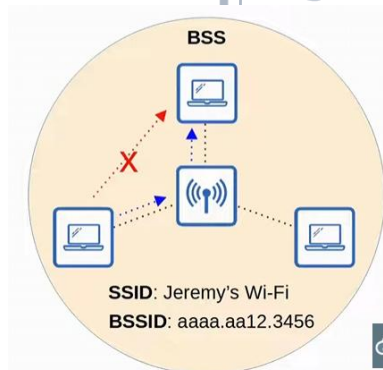
A BSS refers to devices, whereas a BSA refers to a physical area.

Client Communication in a BSS

Clients must communicate through the AP.

Even if another client is within range, traffic still passes through the AP first.

Traffic Flow



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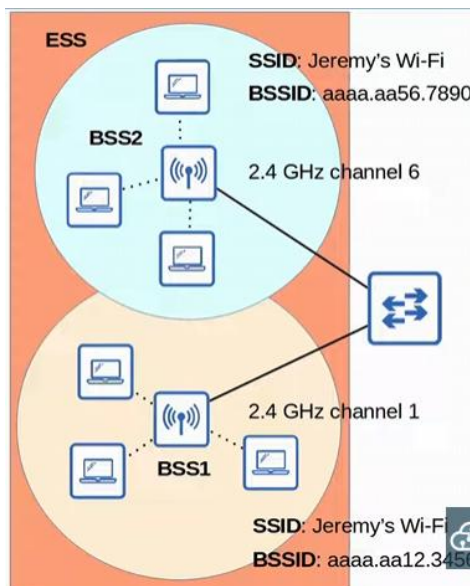
Extended Service Set (ESS)

To create wireless LANs larger than the coverage area of a single AP, an **ESS (Extended Service Set)** is used.

An ESS is the second type of infrastructure service set.

An ESS consists of multiple BSSs connected together by a wired network.

ESS Structure



Each BSS:

1. Uses the same SSID.
2. Uses a unique BSSID.
3. Uses a different channel to avoid interference.

Roaming

Clients can move between APs in an ESS without reconnecting.

This is called roaming.

For roaming to work properly, the BSAs should overlap by approximately: 10% to 15%

Without overlap, connectivity can be lost while moving between APs.

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Mesh Basic Service Set (MBSS)

The final service set type is the MBSS (Mesh Basic Service Set).

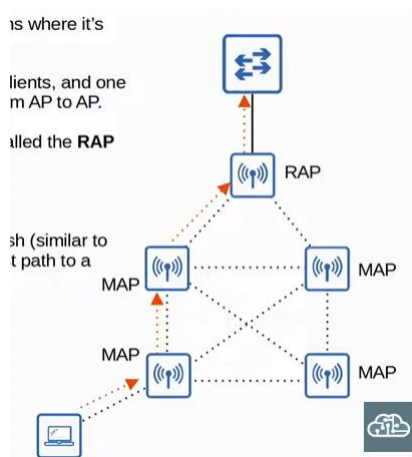
An MBSS is useful when it is difficult to run Ethernet connections to every AP.

Mesh APs use:

1. One radio for wireless clients.
2. One radio for the mesh backhaul network.

The backhaul network carries traffic from AP to AP.

MBSS Structure



RAP-Root Access Point: Connected to the wired network.

MAP-Mesh Access Point: Participates in the mesh network.

Traffic is forwarded through MAPs until it reaches the RAP and then the wired network.

A protocol determines the path traffic should take through the mesh, similar to how routing protocols determine paths in routed networks.

Distribution System (DS)

Most wireless networks are not standalone networks.

Instead, they allow wireless clients to connect to a wired network infrastructure.

In 802.11, the upstream wired network is called the:

DS (Distribution System)

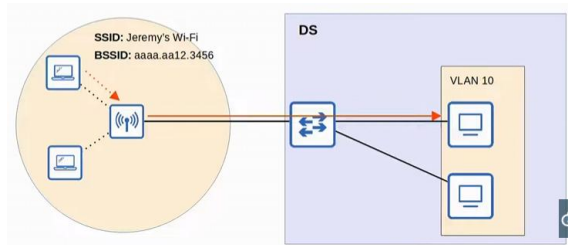
The AP acts as a bridge between the wireless network and the wired network.

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Mapping a Wireless LAN to a VLAN

Each wireless BSS or ESS is mapped to a VLAN in the wired network.

Example:



Wireless hosts associated with the SSID can communicate with hosts in the corresponding VLAN.

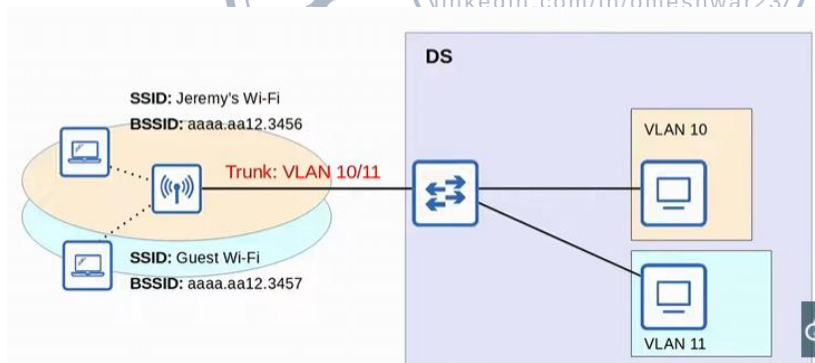
Multiple Wireless LANs on One AP

An AP can provide multiple wireless LANs simultaneously.

Each wireless LAN:

- Has its own SSID.
- Has its own BSSID.
- Maps to its own VLAN.

Example



Notice that the BSSID of the second WLAN is created by incrementing the last digit.

Jeremy's Wi-Fi: 3456

Guest Wi-Fi: 3457

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Repeater Mode

An AP operating in repeater mode extends the range of a BSS.

Repeater Operation

Client ---- Repeater ---- AP ---- Switch

The repeater retransmits signals received from the AP.

Single-Radio Repeater

A repeater with one radio:

- Uses the same channel as the AP.
- Repeats traffic on the same channel.
- Keeps the channel busy.

Result: Effective throughput reduced by 50%

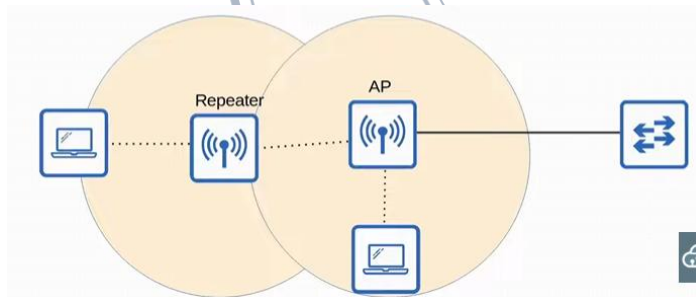
Dual-Radio Repeater

A repeater with two radios can:

Receive ---> Channel A

Transmit ---> Channel B

This avoids the weakness of a single-radio repeater.



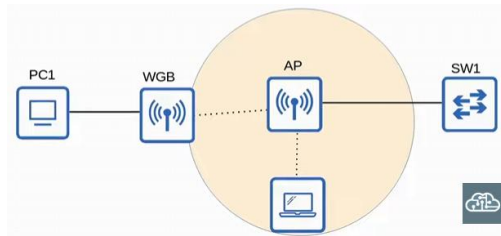
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Workgroup Bridge (WGB)

A **WGB (Workgroup Bridge)** operates as a wireless client of another AP.

It allows wired devices to access a wireless network.

WGB Example



In this example:

- PC1 has no wireless capability.
- PC1 cannot connect directly to SW1.
- PC1 connects to the WGB using Ethernet.
- The WGB connects wirelessly to the AP.

Types of WGB

Universal WGB (uWGB):

An 802.11 standard that allows:

One wired device, to be bridged.

Cisco WGB

Cisco proprietary implementation that allows:

Multiple wired devices to be bridged.

This provides a way for devices without wireless capability to connect to a wireless network.

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Outdoor Bridge Mode

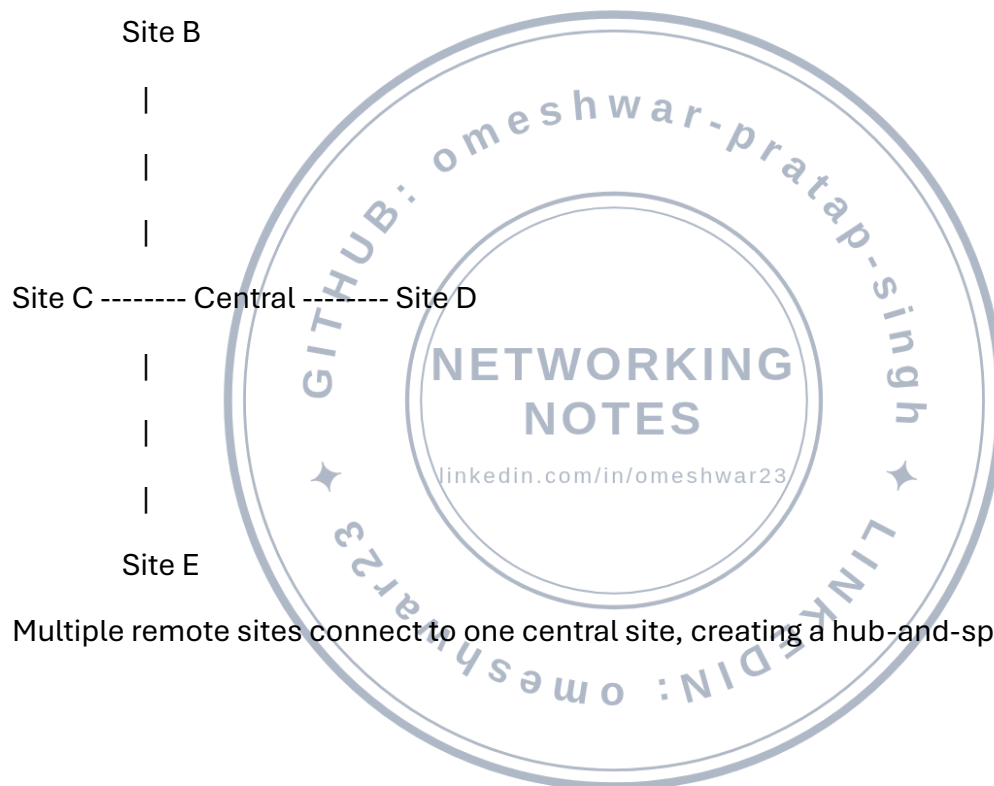
Outdoor bridge mode connects networks over long distances without a physical cable.

Point-to-Point Outdoor Bridge



The APs use specialized antennas that focus signal power in one direction, allowing communication over longer distances.

Point-to-Multipoint Outdoor Bridge



Multiple remote sites connect to one central site, creating a hub-and-spoke topology.

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Summary of Important Terms

Term	Meaning
SSID	Human-readable wireless network name
IBSS	Ad hoc network without an AP
BSS	Infrastructure service set using an AP
BSSID	Unique AP identifier (radio MAC address)
BSA	Physical coverage area of a BSS
ESS	Multiple BSSs connected together
Roaming	Moving between APs without reconnecting
MBSS	Mesh Basic Service Set
RAP	Root Access Point
MAP	Mesh Access Point
DS	Distribution System (wired network)
WGB	Workgroup Bridge
Repeater	Extends BSS coverage
Outdoor Bridge	Long-distance wireless connection between sites

- Wireless LANs are defined in 802.11.
- Operate in half duplex using CSMA/CA
- Wireless signals can be affected by **absorption, reflection, refraction, diffraction, and scattering**.
- Various aspects of waves can be measured, such as **amplitude, frequency, and period**.
- Frequency is measured in **hertz (Hz)**.
- Wireless LANs use two frequency ranges: the **2.4 GHz** band and **5 GHz** band.
 - Wi-Fi 6 (802.11ax) can use the **6 GHz** range too.
- Bands are divided into channels.
- 5 GHz band consists of non-overlapping channels.
- 2.4 GHz band channels overlap. To avoid overlapping, use channels 1, 6, and 11 (in North America).
- 802.11 standards (802.11b, 802.11a, etc) and their frequencies/theoretical max data rates.
- Service sets are groups of wireless devices. Three types:
 - Independent (**IBSS**, also called **ad hoc**)
 - Infrastructure (**BSS, ESS**)
 - *passing between APs in an ESS is called **roaming**.
 - Mesh (**MBSS**)
- Service sets are identified by an **SSID** (non-unique, human-readable) and **BSSID** (unique, MAC address of AP).
- The area around an AP where its signal is usable is called a **BSA**.
- The upstream wired network is called the **DS**.
- When multiple WLANs are used, each is mapped to a separate VLAN on the wired network.
- APs can also operate as a **repeater, workgroup bridge, or outdoor bridge**.

*Although this summarizes the topics in this video, make sure you know the details of each topic that we covered.

